# Exercises Functions

## Exercise

Write a Python program to test whether a number is within 100 of 1000 or 2000. Use a function.
Python abs(x) function:

The function returns the absolute value of a number. The argument may be an integer or a floating point number.

```
def near_thousand(n):
      return ((abs(1000 - n) <= 100) or (abs(2000 - n) <= 100))
print(near_thousand(1000))
print(near_thousand(900))
print(near_thousand(800))
print(near_thousand(2200))
```

## Exercise

Write a Python program to calculate the sum of three given numbers, if the values are equal then return thrice of their sum. Use a function.

```
def sum_thrice(x, y, z):
     sum = x + y + z
     if x == y == z:
      sum = sum * 3
     return sum

print(sum_thrice(1, 2, 3))
print(sum_thrice(3, 3, 3))
```

## Exercise

Write a Python program to get a new string from a given string where "Is" has been added to the front. If the given string already begins with "Is" then return the string unchanged. Use a function.

```
def new_string(str):
  if len(str) >= 2 and str[:2] == "Is":
    return str
  return "Is" + str

print(new_string("Array"))
print(new_string("IsEmpty"))
```

## Exercise

Write a Python program to count the number 4 in a given list. Use a function.

```
def list_count_4(nums):
  count = 0
  for num in nums:
    if num == 4:
      count = count + 1

  return count
print(list_count_4([1, 4, 6, 7, 4]))
print(list_count_4([1, 4, 6, 4, 7, 4]))
```

## Exercise

Write a Python program to test whether a passed letter is a vowel or not.

```
def is_vowel(char):
    all_vowels = 'aeiou'
    return char in all_vowels
print(is_vowel('c'))
print(is_vowel('e'))
```

## Exercise

Write a Python program to print the even numbers from a given list.

```
def is_even_num(l):
    enum = []
    for n in l:
        if n % 2 == 0:
            enum.append(n)
    return enum
print(is_even_num([1, 2, 3, 4, 5, 6, 7, 8, 9]))
```

## Exercise

Write a Python function to find the Max of three numbers.

```
def max_of_two( x, y ):
    if x > y:
        return x
    return y
def max_of_three( x, y, z ):
    return max_of_two( x, max_of_two( y, z ) )
print(max_of_three(3, 6, -5))
```